

Chapter 2

The Project Management and Software Engineering Process context:

The Organizational Context: Strategy, Structure, and Culture

Projects and Organizational Strategy

Strategic management: the science of formulating, implementing, and evaluating **cross-functional decisions** that enable an **organization** to achieve its **objectives**.

Consists of:

- Developing vision and mission statements
- Formulating, implementing, and evaluating
- Making cross-functional decisions
- Achieving objectives

Table 2.1 Projects Reflect Strategy

Strategy	Project
Technical or operating initiatives (such as new distribution strategies or decentralized plant operations)	Construction of new plants or modernization of facilities
Development of products for greater market penetration and acceptance	New product development projects
New business processes for greater streamlining and efficiency	Reengineering projects
Changes in strategic direction or product portfolio reconfiguration	New product lines
Creation of new strategic alliances	Negotiation with supply chain members (including suppliers and distributors)
Matching or improving on competitors' products and services	Reverse engineering projects
Improvement of cross-organizational communication and efficiency in supply chain relationships	Enterprise IT efforts
Promotion of cross-functional interaction, streamlining of new product or service introduction, and improvement of departmental coordination	Concurrent engineering projects



What went wrong with the A380? The aircraft itself features cutting edge technology, has modern conveniences (Emirates offers its first-class passengers on-board showers and a fully stocked bar and lounge area), and is popular with travelers. Unfortunately, its design has also led to several ancillary problems. For example, because the airplane is so large, it requires a longer-than-average runway for takeoffs and landings. Furthermore, the two-decker configuration requires airports to redesign and install special jet bridges and entrance/exit gates to allow for double lines of passengers boarding and leaving the aircraft. These redesigns have proven to be unpopular with many airports that refuse to make the necessary modifications. When the A380 was first introduced in the United States, only eight airports had taxiways wide enough and parking ramps large enough to accommodate the plane. Interestingly, the sheer size of the aircraft also creates operational hazards for other flights landing or taking off behind it. As one writer put it, “The A380 is basically a flying weather system, with its 261-foot wings throwing off hurricane-strength winds. Under current international guidelines, smaller aircraft need a seven-mile gap between themselves and the A380.” By comparison, some airports that handle smaller jets can space them as little as 2.5 miles apart. The arrival and departure of multiple A380s can slow flight operations at some of the busiest airports in the country.

What went wrong for the A380? Much of the blame rests on the strategic decisions taken by Airbus and its view of the future of global aviation. At the beginning of the millennium, Airbus and Boeing looked where their business was heading and saw similar facts: air traffic doubling every 15 years, including estimates that the number of travelers would hit four billion by 2020 — and came to radically different conclusions for the future. For Airbus there was need for much bigger planes for sustainable growth at congested airports. For Boeing, medium-sized wide-body jets best matched future globalization pressures and anticipated higher frequency of travel among major and multiple growing cities.

Projects and Organizational Strategy

Strategic management consists of the following elements

1. **Developing vision statements and mission statements.** Vision and mission statements establish a sense of what the organization hopes to accomplish or what top managers hope it will become at some point in the future.
2. **Formulating, implementing, and evaluating.** Projects, as the key ingredients in strategy implementation, play a crucial role in the basic process model of strategic management. A firm devotes significant time and resources to evaluating its business opportunities through developing a corporate vision or mission, assessing internal strengths and weaknesses as well as external opportunities and threats, establishing long-range objectives, and generating and selecting among various strategic alternatives
3. **Making cross-functional decisions.**
4. **Achieving objectives.**

TOWS Matrix

	External Opportunities (O)	External Threats (T)
Internal Strengths (S)	SO "Maxi-Maxi" Strategy Develop projects that use strengths to maximize opportunities	ST "Maxi-Mini" Strategy Develop projects that use strengths to minimize threats
Internal Weaknesses (W)	WO "Mini-Maxi" Strategy Develop projects that minimize weaknesses by taking advantage of opportunities	WT "Mini-Mini" Strategy Develop projects that minimize weaknesses and avoid threats

Stakeholder Management

Organizations and project teams cannot operate in ways that ignore the external effects of their decisions. One way to understand the relationship of project managers and their projects to the rest of the organization is through employing stakeholder analysis.

Stakeholder analysis is a useful tool for demonstrating some of the seemingly irresolvable conflicts that occur through the planned creation and introduction of new projects.

Project stakeholders are defined as all individuals or groups who have an active stake in the project and can potentially impact, either positively or negatively, its development.

Identifying Project Stakeholders

Internal Stakeholders

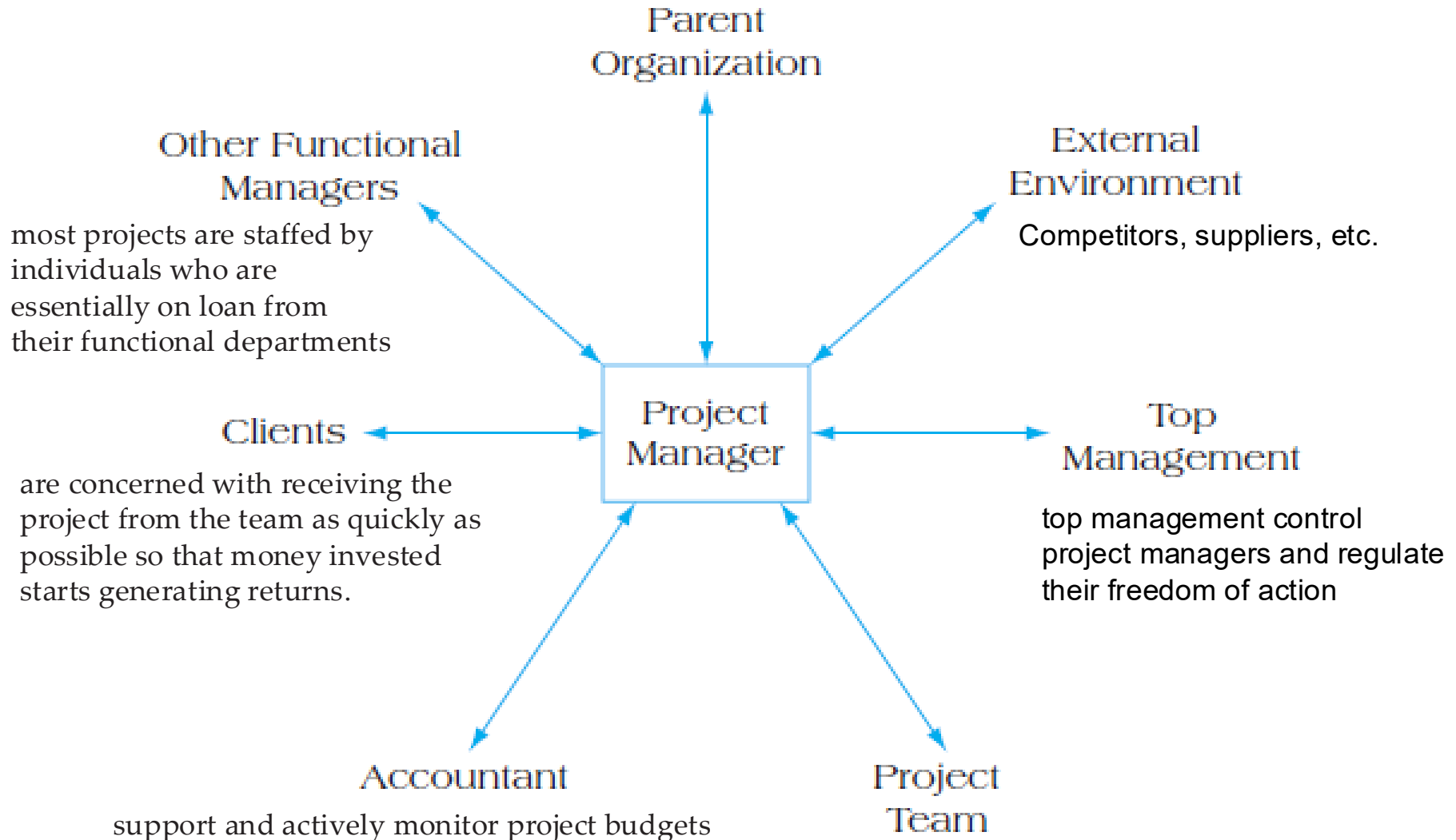
- Top management
- Accountant
- Other functional managers
- Project team members

External Stakeholders

- Clients
- Competitors
- Suppliers
- Environmental, political, consumer, and other intervenor groups

- Hydraulic fracturing technology has been widely embraced in USA which led to going from being a net importer of natural gas in 2012 to a net exporter in 2016. Environmental groups, however, continue to oppose these steps, using litigation and political lobbying to resist the development of these energy sources. They were initially successful in halting completion of the Dakota Access Pipeline project and have delayed for years the development of the 1,700-mile-long Keystone XL oil pipeline from Canada's oil sands region to refineries in Texas. These types of external stakeholders are called **intervenor groups** (possessing the power to effectively intervene and disrupt the project).

Figure 2.3 Project Stakeholder Relationships



Organizational Structure

Consists of three key elements:

1. Designates formal reporting relationships, Who reports to whom?
 - number of levels in the hierarchy
 - span of control
2. Identifies groupings of
 - individuals into departments
 - departments into the total organization
3. Design of systems to ensure
 - effective communication
 - coordination
 - integration of effort across departments

Forms of Organization Structure

- **Functional organizations**—group people performing similar activities into **departments**
- **Project organizations**—group people into **project teams** on temporary assignments
- **Matrix organizations**—create a dual hierarchy in which **functions and projects** have equal prominence

Figure 2.4 Example of a Functional Organizational Structure

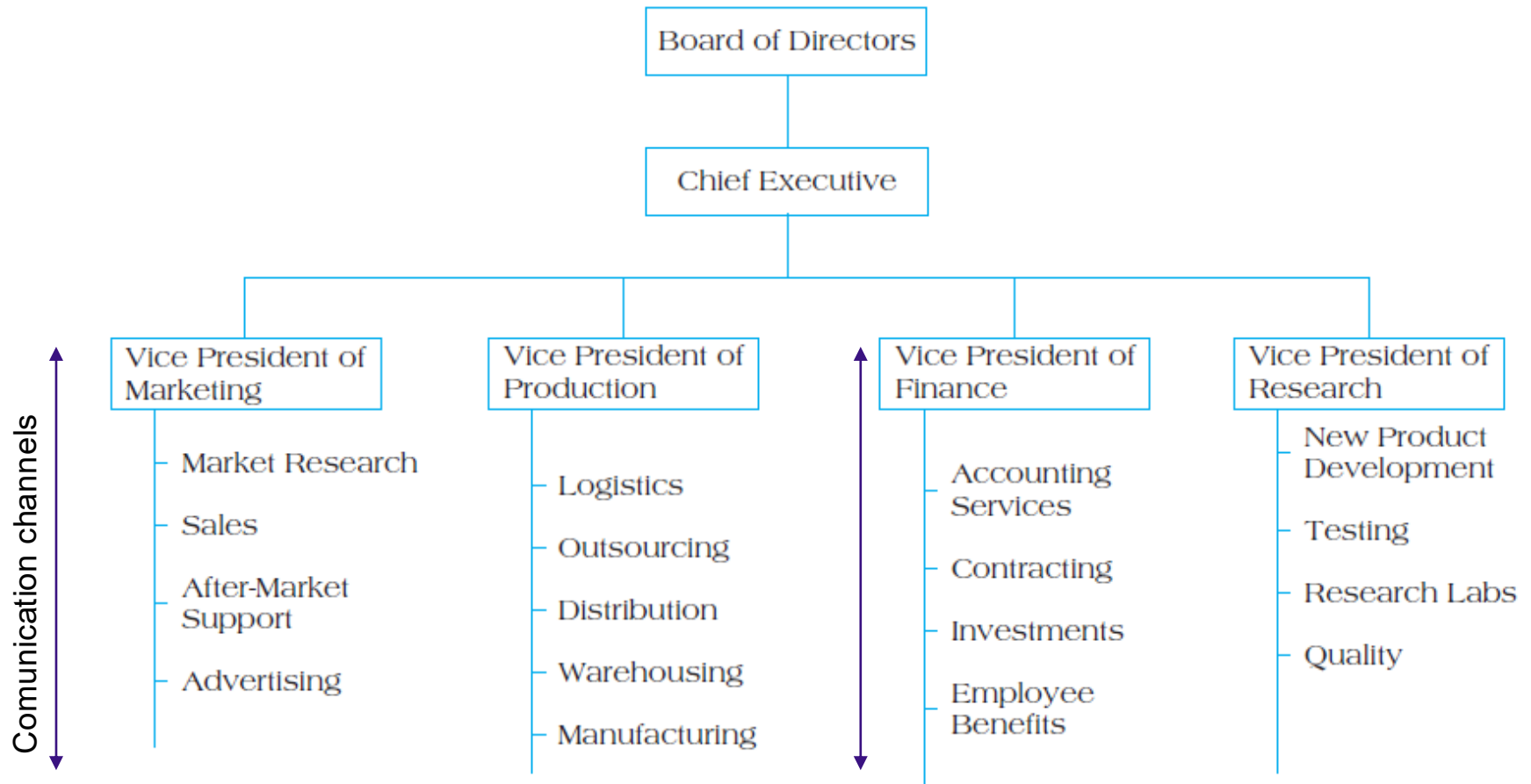
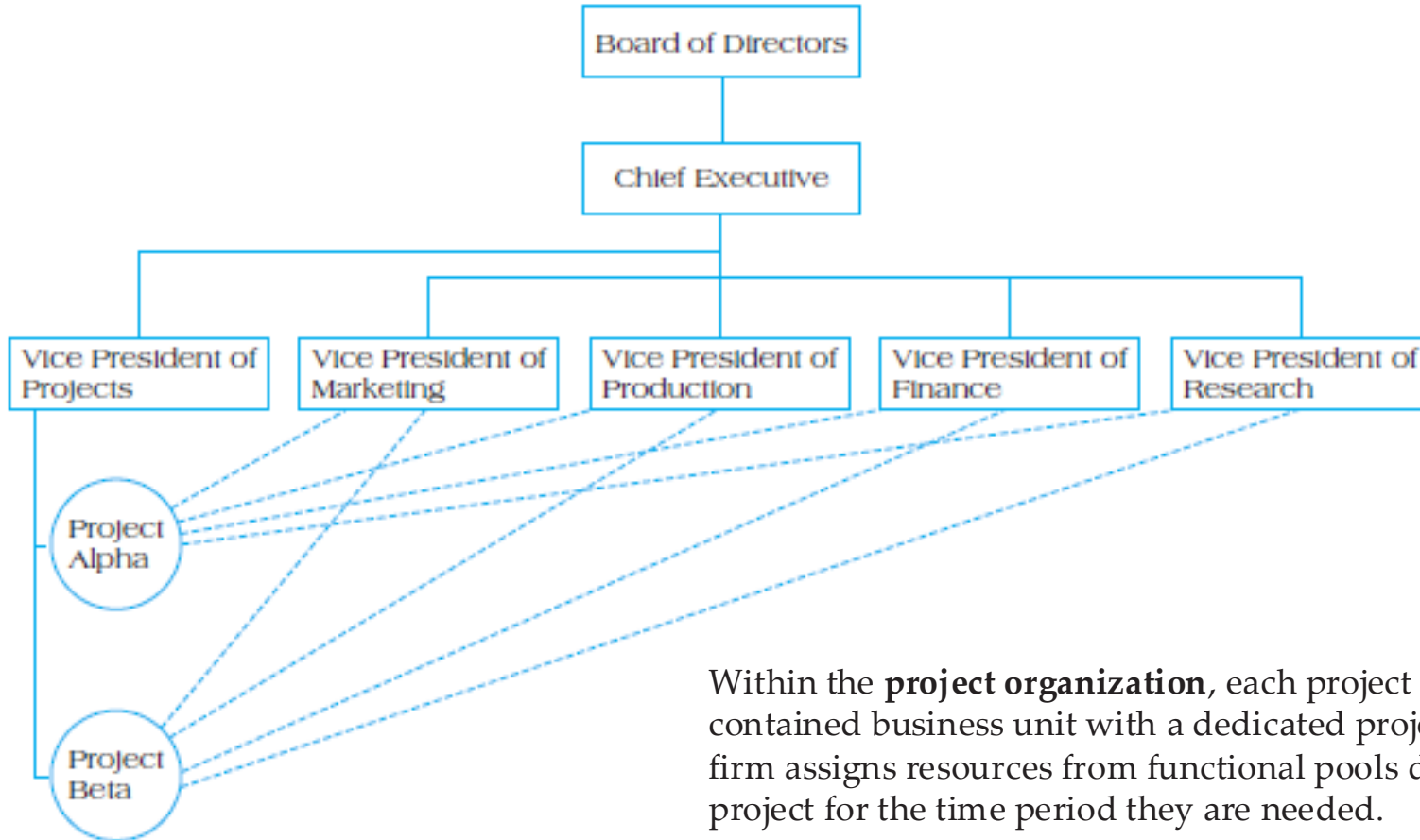


Table 2.2 Strengths and Weaknesses of Functional Structures

Strengths for Project Management	Weaknesses for Project Management
1. Projects developed within basic functional structure require no disruption or change to firm's design. 2. Enables development of in-depth knowledge and intellectual capital. 3. Allows for standard career paths.	1. Functional <u>siloing</u> (unwillingness to consider alternative viewpoints, collaborate with other groups, etc.) makes it difficult to achieve cross-functional cooperation. 2. Lack of customer focus. 3. Longer time to complete projects due to structural problems, slower communication, etc. 4. Varying interest or commitment.

Project Organizational Structure



Within the **project organization**, each project is a self-contained business unit with a dedicated project team. The firm assigns resources from functional pools directly to the project for the time period they are needed.

Project organizations are those that are set up with their exclusive focus aimed at running projects. Construction companies, large manufacturers such as Boeing or Airbus, pharmaceutical firms, and many software consulting and research and development organizations are organized as pure project organizations.

Table 2.3 Strengths and Weaknesses of Project Structures

Strengths for Project Management	Weaknesses for Project Management
1. Project manager sole authority	1. Expensive to set up and maintain teams
2. Improved communication	2. Chance of loyalty to the project rather than the firm
3. Effective decision making	3. Difficult to maintain a pooled supply of intellectual capital (contractors hired temporarily for the project then leave the organization)
4. Creation of project management experts	
5. Rapid response to market opportunities	4. Team member concern about future once project ends

Figure 2.7 Example of a Matrix Organizational Structure

The vice president of projects controls the activities of the project managers under his authority. They, however, must work closely with functional departments to staff their project teams through loans of personnel from each functional group

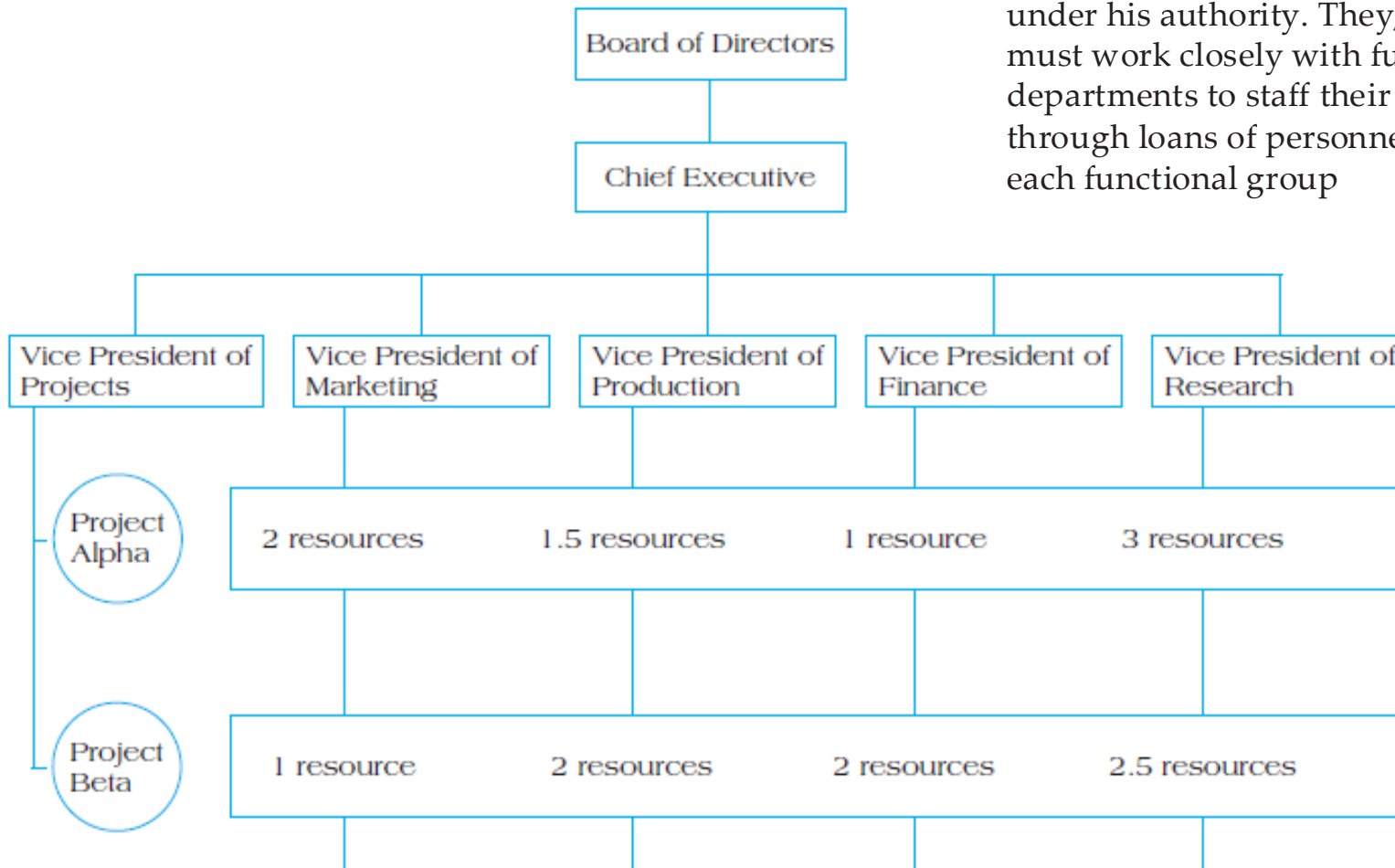


Table 2.4 Strengths and Weaknesses of Matrix Structures

Strengths for Project Management	Weaknesses for Project Management
1. Suited to dynamic environments	1. Dual hierarchies mean two bosses
2. Equal emphasis on project management and functional efficiency	2. Negotiation required in order to share resources
3. Promotes coordination across functional units	3. Workers caught between competing project and functional demands
4. Maximizes scarce resources	

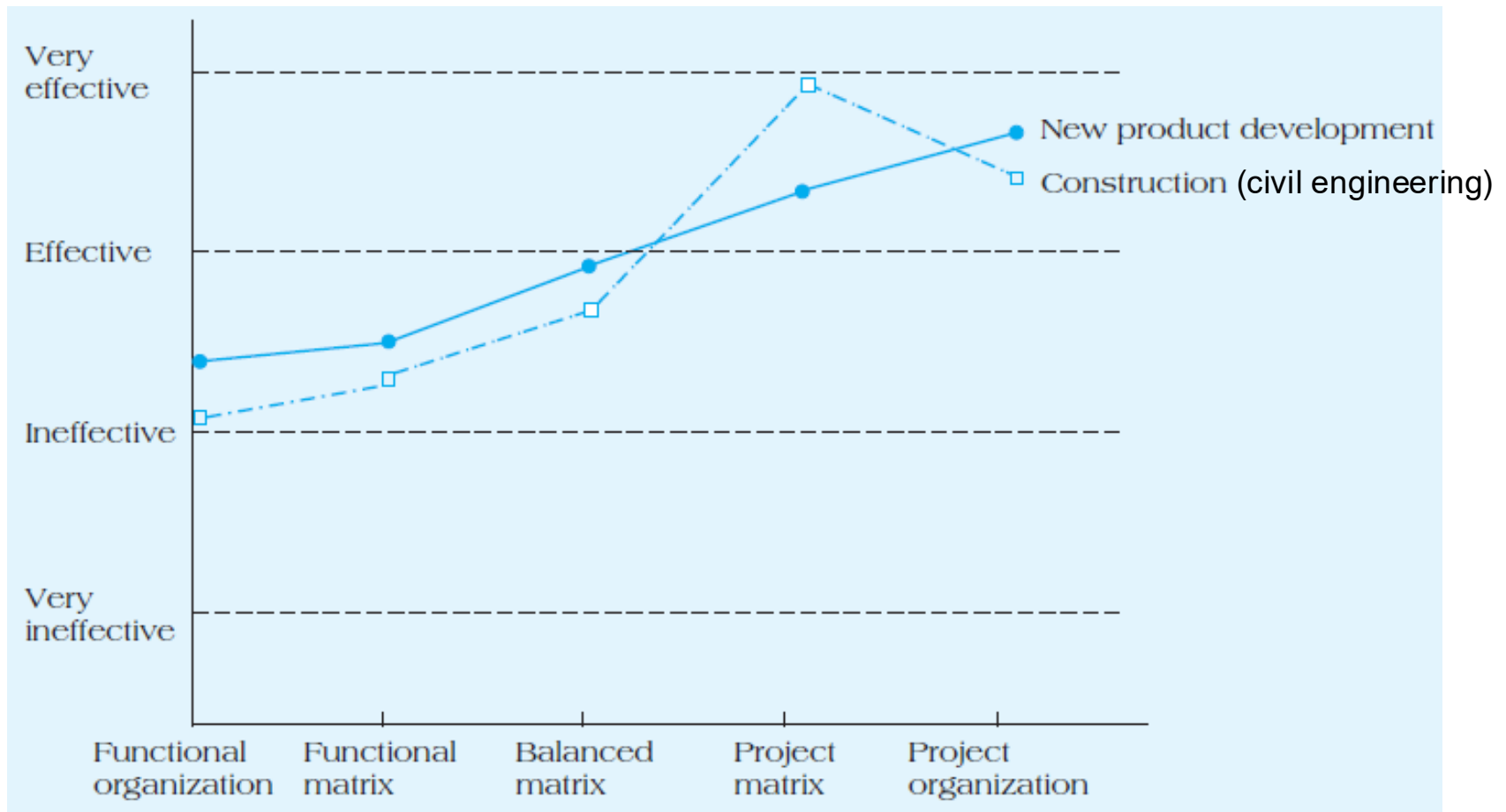
Heavyweight Project Organizations

Organizations can sometimes gain tremendous benefit from creating a **fully dedicated project organization**.

Lockheed Corporation's "Skunk works" model (small autonomous team of highly skilled hand-picked individuals given complete freedom, the project manager is given full authority, status, and responsibility)

- Project manager authority expanded
- Functional alignment abandoned in favor of market opportunism, direct reporting to the highest executive levels
- Focus on external customer

Figure 2.8 Managers' Perceptions of Effectiveness of Various Structures on Project Success (survey study)



Project Management Offices

Centralized units that oversee or improve the management of projects

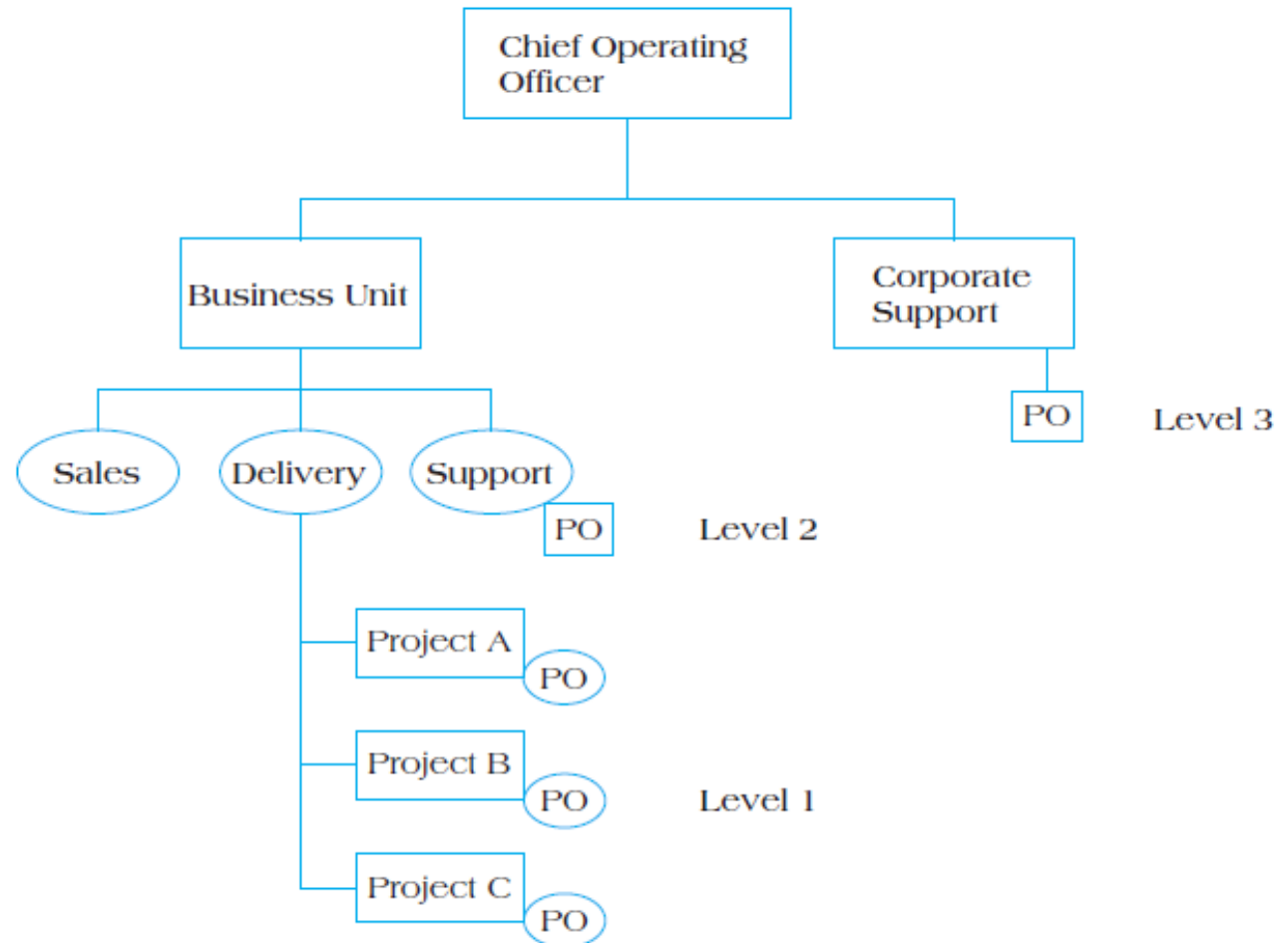
Resource centers for:

- Technical details
- Expertise
- Repository
- Center for excellence

Using a PMO as a resource center shifts some of the burden for PM activities (activity scheduling, resource allocation, monitoring, control processes, etc.) from the project manager to a support staff that is dedicated to providing this assistance

Alternative Levels of Project Offices

The PMO may be situated at a corporate level (Level 3), where it serves an overall corporate support function. It can be placed at a lower functional level (Level 2), where it serves the needs within a specific business unit. Finally, the PMO can be decentralized down to the actual project level (Level 1), where it offers direct support for each project.



Forms of PMOs and Control

Three forms of PMOs, varying with degrees of control and influence include:

- **Supportive**, low control; consultative and provide PM resources and training
- **Controlling**, moderate control; requires compliance to adopted PM standards/processes
- **Directive**, high control; directly manages projects

Models of PMOs

In addition to the forms of PMO and varying levels of control, there are three models of PMOs with various purposes for companies:

- **Weather station**—the PMO is typically used only as a monitoring and tracking device (management wants to keep an eye on the status of the projects without directly attempting to influence or control them)
- **Control tower**—project management is a skill to be protected and supported (focus on improving methodology and practices and help PM apply them)
- **Resource pool**—maintain and provide a cadre of skilled project professionals (supplying project managers and other skilled professionals to the company's projects)

Organizational Culture

- The third key contextual variable in how projects are managed effectively is that of organizational culture
- Definition of culture: “**organization’s culture** refers to the **unwritten rules of behavior**, or norms that are used to shape and guide behavior, that are shared by some subset of organizational members and that are taught to all new members of the company”
 - Unwritten (are often not written down)
 - Rules of behavior (guidelines as to how best to react to these events)
 - Held by some subset of the organization (may or may not be companywide)
 - Taught to all new members (New members of an organization pick up the behaviors as they observe others engaging in them)
- Ex: Company X has a strong supportive culture that emphasizes and rewards positive collaboration between functional groups.

Key Factors That Affect Culture Development

Among the key factors that affect the development of a culture are

- Technology
- Environment
- Geographical location
- Reward systems
- Rules and procedures
- Key organizational members
- Critical incidents (moments when real culture becomes visible)

“ General Electric’s Jet Engine Division and Rolls-Royce share many features, including product lines. Both produce jet engines for the commercial and defense aircraft industries. However, GE prides itself on its competitive, high-pressure culture that rewards aggressiveness and high commitment but also has a high burnout rate among engineers and mid-level managers. Rolls-Royce, on the other hand, represents an example of a much more paternalistic culture that rewards loyalty and long job tenure. “

Organizational Culture: Effects on Project Management

Culture can affect project management in at least four ways:

- **Departmental interaction:** cultures that favor active cooperation between functional groups and new projects are much more successful than those that adopt a disinterested or even adversarial relationship
- **Employee commitment to goals:** A culture that promotes employee commitment and self-sacrifice through working extra hours or on multiple tasks is much more successful than a culture in which the unwritten rules seem to imply that there is nothing wrong with simply going through the motions, provided you don't get caught.
- **Project planning:** activity estimation is an imprecise process, it is common for team members to give wider estimates to give themselves as much time as possible. These people are often responding to a culture that reinforces the idea that it is better to engage in poor estimation and project planning than to be late. In the contrary: Culture of trust -> honest assessments.
- **Performance evaluation:** if organizations tacitly support caution and playing it safe, their project management approaches will equally reflect this principle